fordable, inclusive and population-scale solutions across sectors.

Compute Layer: The Chips Behind the Brain

The third layer of the AI stack is the compute needed for training and inference. This compute is powered by advanced chips. Modern AI relies on powerful processors such as NVIDIA's Blackwell GPUs, Google's TPUs, NPUs, and other server-grade chips that power large-scale training and inference.

Under the IndiaAI mission, we are supporting Indian researchers, startups and academia in training their AI models. Around 40,000 GPUs are being provided at a subsidised average rate of Rs 65 per hour, which is one-third of the global average cost. In addition, custom chips are being developed to meet specific training and inference requirements. In India too, several startups are designing specialised AI chips, strengthening domestic capabilities in AI compute.

This effort is being supported by India's growing semiconductor ecosystem. Ten semiconductor projects have already been approved, including two fabs (fabrication) and eight ATMP (assembly, testing, marking and packaging) units. These projects will create chip development capabilities in India and eventually support the compute needs of India's AI and digital infrastructure in future.

Data Centres: AI Infrastructure Layer

Data centre and network infrastructure is the fourth layer of the AI stack. Under the prime minister's forward-looking leadership, India has covered more than 85 per cent of the country with 5G services. Optical fiber cables criss-cross the country, providing network efficiencies. And now close to Rs 6 lakh crore is being invested in data centres to create the brain power that will propel us in the Fifth Industrial Revolution.

Innovation is also reshaping how data centres are built and operated. New approaches are improving cooling systems, enhancing water efficiency and reducing overall energy consumption. Expanding data centre capacity within India also reduces dependence on foreign digital infrastructure. It ensures that India's AI models, datasets and innovation pipelines remain within the country's digital jurisdiction.

Global technology leaders such as Google, Microsoft and Amazon have already announced major long-term investments in AI and data centre infrastructure in the country. Indian companies are also investing heavily.

> EXPANDING DATA CENTRE CAPACITY ENSURES THAT **INDIA'S AI MODELS, DATASETS AND INNOVATION PIPELINES** REMAIN WITHIN THE COUNTRY'S DIGITAL JURISDICTION

These reflect confidence in India's digital future. The growth of data centres also generates high-value employment and supports strong local innovation ecosystems.

Energy: Fifth Layer in the AI Stack

Energy is the final layer of the AI stack. AI data centres are energy-intensive. As AI adoption grows, the need for more data centres and high-capacity power will increase. The electricity demand for data centres is expected to increase significantly.

This makes sustainable and reliable power solutions critical. Solar and wind energy are central to India's clean energy transition. However, they are intermittent and cannot by themselves meet the round-the-clock power needs of large AI systems and data centres.

Nuclear energy, therefore, becomes an important energy source, providing clean, stable continuous power for AI infrastructure. Prime Minister Narendra Modi foresaw this challenge. The Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Act, recently passed by Parliament, positions nuclear power as a stable and round-the-clock source of clean energy for AI and data centres.

As research in nuclear technology grows, we can expect small modular and micro reactors that will generate 15-50 megawatts of power and be deployed in compact, container-sized units. These will enable the supply of clean and reliable energy even while maintaining high safety standards. This makes nuclear power a practical on-site solution for energy-intensive digital infrastructure.

The SHANTI Act also facilitates public-private partnerships and foreign investment in India's nuclear sector. This will bring in capital, technology and expertise in the nuclear energy sector and ultimately help the development of AI and data centres.

AI for Humanity

We are building the full AI stack, step by step. It is to make AI work at scale, rooted in India's realities, affordable for innovators, and delivering real impact in people's lives. Each layer of the stack, from applications and models to compute, infrastructure and energy, is designed to remove barriers and enable widespread adoption.

As PM Modi said at the Paris AI Action Summit last year, "AI is writing the code for humanity in this century." We are building each layer of this stack with a sense of responsibility towards our people. We are making sure that the code so written fits our context, reflects our values, and actually solves real problems of people. ■

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